Tejas Session 2 Operators

Tuesday, July 7, 2026

Home Work:-

x = input ( ) → user
y =

print (x+y)
print (x-y)
print (x*y)
print (x/y)

Data Types

variables ↑

What kind of data it stores → data type

Imagine a bank account.

A bank account contains different kinds of information:

- Account Holder Name → `"Shanmuka"`
- Account Balance → `50000.75`
- Account Active → `True`
- Transaction IDs → `[101, 102, 103]` → Collection

All these values are different.

Python needs to know:

- Which value is text?
- Which value is a number?
- Which value is a yes/no answer?
- Which value is a collection?

This is where **Data Types** become important.

Data Types

Single Value

Collection Data Types

These store one value at a time.
Examples:

- int ✓ → age → 12
- float ✓ → wt → 42.35
- str ✓ → name → "Tejas"
- bool ✓ → isactive → false

These store multiple values together.

Examples:

- list
- tuple
- set
- dict

that can change

leave

single

Data Type	Example	Mutable/Immutable	Ordered/Unordered	Real-Life Example
int	100	Immutable	N/A	Roll Number
float	99.5	Immutable	N/A	Product Price
str	"Python"	Immutable	Ordered	Student Name

Sing

int	100	Immutable	N/A	Roll Number
float	99.5	Immutable	N/A	Product Price
str	"Python"	Immutable	Ordered	Student Name
bool	True	Immutable	N/A	Login Status
list	[1,2,3]	Mutable	Ordered	Shopping Cart
tuple	(1,2,3)	Immutable	Ordered	Coordinates
set	{1,2,3}	Mutable	Unordered	Unique IDs
dict	{"name":"Rahul"}	Mutable	Insertion-Ordered	Student Record

Collectn

```
name="Azaan"
age=24
hobbies=["VOlleyball","Programming"]

print(type(name))
print(type(age))
print(type(hobbies))
```

Operators

operators → $\rightarrow$ operators.

$x \boxed{+} y$

$x + y$

$x - y$

$x * y$

Types of operators

round off value

10//3 = 3

10/3 = 3.33

1) Arithmetic Operators :-

+ → Addition ✓

− → subtraction ✓

// → floor division

% → modulus ✗

** → Exponentiation ✗

* → multiplication ✓

/ → Division

division

Modulus (%) :-

It provides remainder

Ex- 10 % 3 = 1

10 % 5 = 0

$$3\overline{)10}^{\,3}$$
−9
1

$$5\overline{)10}^{\,2}$$
−10
0

Note :- a % b

If a < b
ans = a

2 % 5 = 2

Exponentiation :- (Power)

3 ** 2 = 9

4 ** 3 = 64

Comments :-

ignore by python

This is a comment

2) Assignment operator :-

+=
-=
*=
%=
**=
//=

$\boxed{x = x \text{ operator value}}$

x = 10
x += 1
x = x + 1 = 11

x = 10
x *= 10
print(x)
⇒ x = x * 10 = 100

x = 10
print(x)

x += 10
print (x)

O/P = 20

x = x + 10 = 20

x
10

x = x + 10
(previous value)

x = 5
x += 4
print (x) O/P = 9

x -= 1 → x = x - 1 = 9

x += 4
x = x + 4 = 9

x
9

3) Relation operator :-

True | false

==
>
<

== Equal To ✓
> Greater Than ✓
< Lesser Than ✓
>= Greater Than Equal To ✓

x = 10 y = 10

x = 2 y = 10

a = 10 b = 11

10 > 11 ✗ false

$<$

$>=$

$<=$

$!=$

<	Lesser Than
>=	Greater Than Equal To
<=	Lesser Than Equal To
!=	Not Equal To

a=10 b=11

a>b → 10>11 × false

x=10 y=2

x≥2 → $\underbrace{x>2 \text{ or } x==2}$ → True

!= → not equals

x=10 y=5

x!=y → True

x=10 y=10

x!=y → false

4) Logical Operators:

AND →

Both condⁿ → should be True

OR

Any one need to be true

x=10

y=5

z=3

print($\underbrace{x>y \text{ and } y>z}$) → True ↳ Both true

print(x>y and y<z) → false

x=10

y=5

z=3

print($\underbrace{x>y \text{ or } y>z}$) → True — either one true

print(x>y or y<z) → True

Operator	Meaning	int	float	complex	bool	string
+	Addition / Concatenation	print(10 + 5) Output: 15	print(10.5 + 2.5) Output: 13.0	print((2+3j) + (1+2j)) Output: (3+5j)	print(True + False) Output: 1	print("Hello " + "Python") Output: Hello Python
-	Subtraction	print(10 - 5) Output: 5	print(10.5 - 2.5) Output: 8.0	print((5+4j) - (2+1j)) Output: (3+3j)	print(True - False) Output: 1	print("Hello" - "H") Output: TypeError
*	Multiplication / Repetition	print(10 * 5) Output: 50	print(2.5 * 4) Output: 10.0	print((2+3j) * 2) Output: (4+6j)	print(True * 5) Output: 5	print("Hi " * 3) Output: Hi Hi Hi
/	Division	print(10 / 5) Output: 2.0	print(7.5 / 2.5) Output: 3.0	print((6+4j) / 2) Output: (3+2j)	print(True / 2) Output: 0.5	print("Hello" / 2) Output: TypeError
//	Floor Division	print(10 // 3) Output: 3	print(10.5 // 3) Output: 3.0	print((6+4j) // 2) Output: TypeError	print(True // 2) Output: 0	print("Hello" // 2) Output: TypeError
%	Modulus	print(10 % 3) Output: 1	print(10.5 % 3) Output: 1.5	print((6+4j) % 2) Output: TypeError	print(True % 2) Output: 1	print("Hello" % 2) Output: TypeError
**	Exponent	print(2 ** 3) Output: 8	print(2.5 ** 2) Output: 6.25	print((2+3j) ** 2) Output: (-5+12j)	print(True ** 3) Output: 1	print("Hello" ** 2) Output: TypeError